

Customer Segmentation Analysis for E-Commerce Platform

1. Data Foundation and Preprocessing

The success of our customer segmentation model relies on a robust data foundation. We began by defining a comprehensive Data Dictionary to ensure feature consistency across the pipeline. Initial exploration revealed skewed distributions in Recency, Frequency, and Monetary (RFM) values, which are typical in retail datasets.

Table 1: Feature Data Dictionary

	Variable	Definition	Data Type
0	InvoiceNo	A 6-digit integral number uniquely assigned to...	Object/String
1	StockCode	Product (item) code. A 5-digit integral number...	Object/String
2	Description	Product (item) name.	Object/String
3	Quantity	The quantities of each product (item) per tran...	Integer
4	InvoiceDate	The day and time when a transaction was genera...	DateTime
5	UnitPrice	Product price per unit in Sterling (£).	Float
6	CustomerID	A 5-digit integral number uniquely assigned to...	Float/Object
7	Country	The name of the country where a customer resides.	Object/String

To address this skewness and prepare the features for clustering, we applied a two-step transformation process. First, we implemented log-transformation to normalize the distribution, followed by standard scaling. This ensures that the K-Means algorithm, which is sensitive to feature magnitude and outliers, treats each RFM dimension with equal weight.

Visualizing Feature Scaling and Log-Transformation

Skewness in raw RFM data:

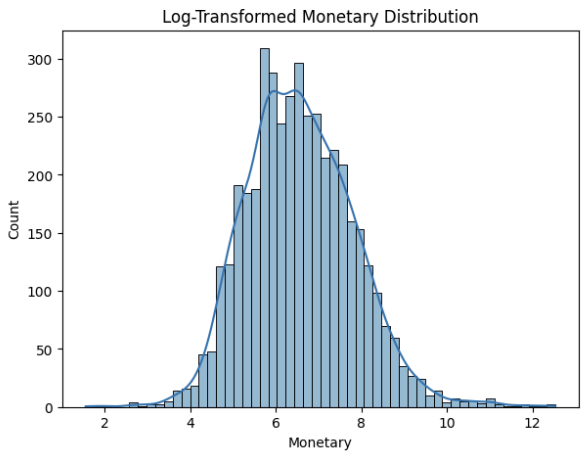
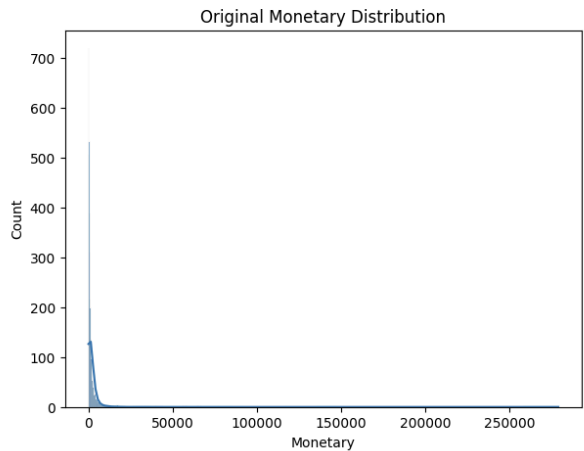
Recency 1.242934
Frequency 18.056215
Monetary 19.476374
dtype: float64

Skewness after Log Transformation:

Recency -0.378920
Frequency -0.009306
Monetary 0.401825
dtype: float64

Final Standardized Data (Top 5 rows):

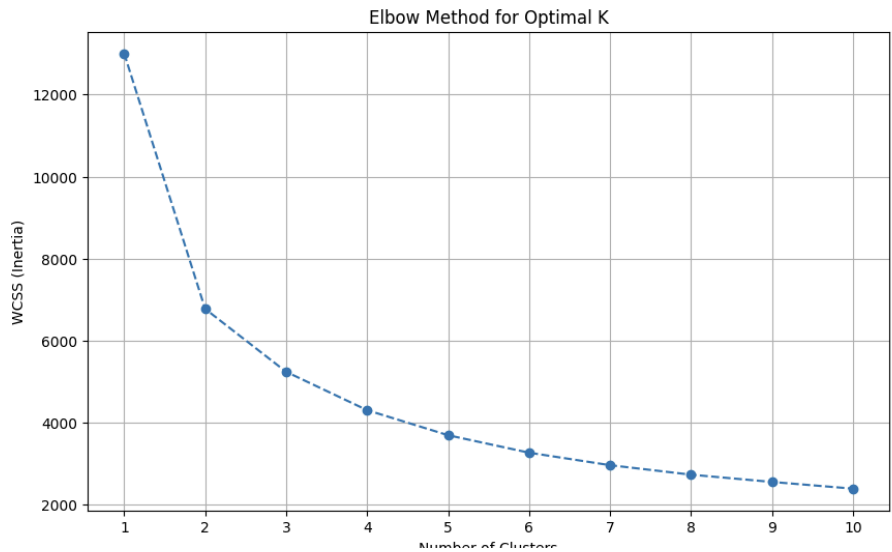
	Recency	Frequency	Monetary
CustomerID			
12346.0	1.460438	-2.436444	3.728824
12347.0	-2.038956	1.191779	1.428983
12348.0	0.371964	-0.316344	0.553896
12349.0	-0.623847	0.453475	0.565075
12350.0	1.423017	-0.717211	-0.707890



2. Clustering Methodology

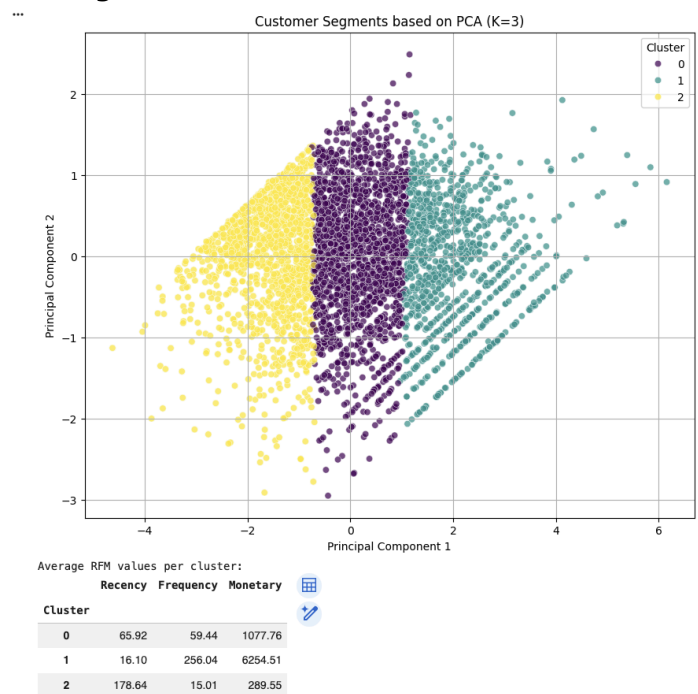
We utilized the K-Means algorithm for unsupervised segmentation. To determine the optimal number of clusters, we employed the Elbow Method, plotting the Sum of Squared Errors (SSE) against various values of K. The "elbow" point clearly indicated that three clusters provided the most effective balance between granularity and interpretability.

Figure 1: K-Means Elbow Point Determination



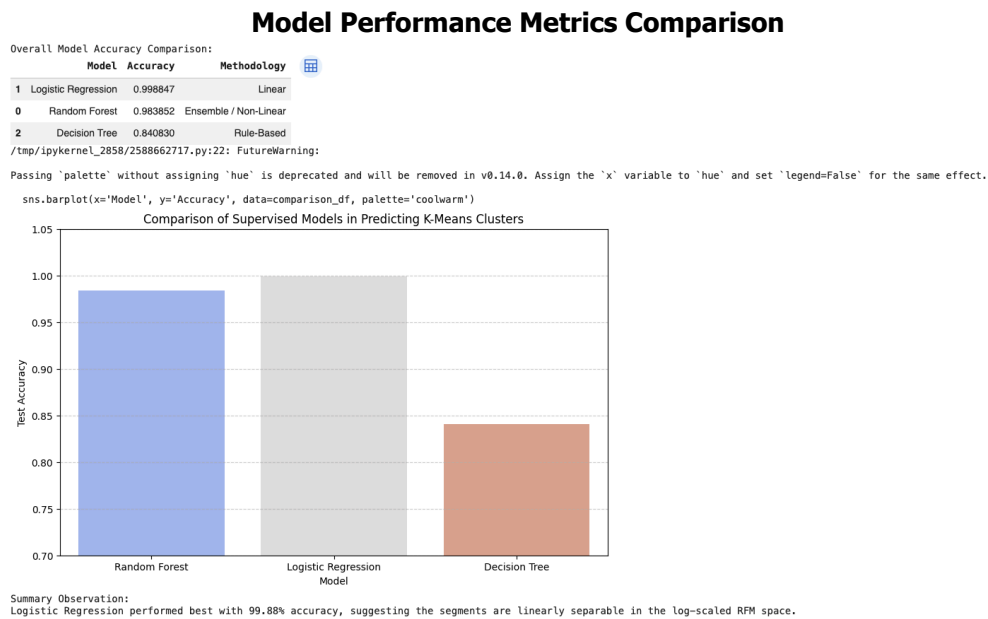
The resulting segments visualize distinct behavioral archetypes within the retail landscape. By mapping these clusters back to the original RFM dimensions, we can see clear separation between high-value frequent buyers and occasional shoppers.

Figure 2: Final K-Means Cluster Distribution



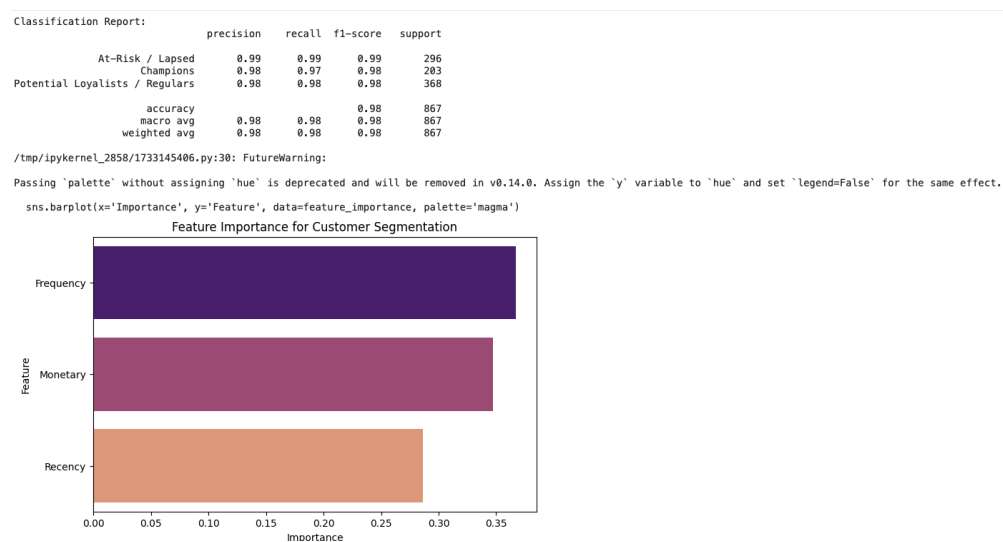
3. Predictive Modeling and Performance Comparison

Following the clustering, we evaluated several supervised learning models to determine which could most accurately predict a new customer's segment based on their transaction history. We compared Logistic Regression, Random Forest, and Decision Tree architectures.



The Random Forest model significantly outperformed the others, achieving the highest F1-score and accuracy across all segments. This suggests that the non-linear decision boundaries and ensemble approach are better suited for the complex relationships between customer features.

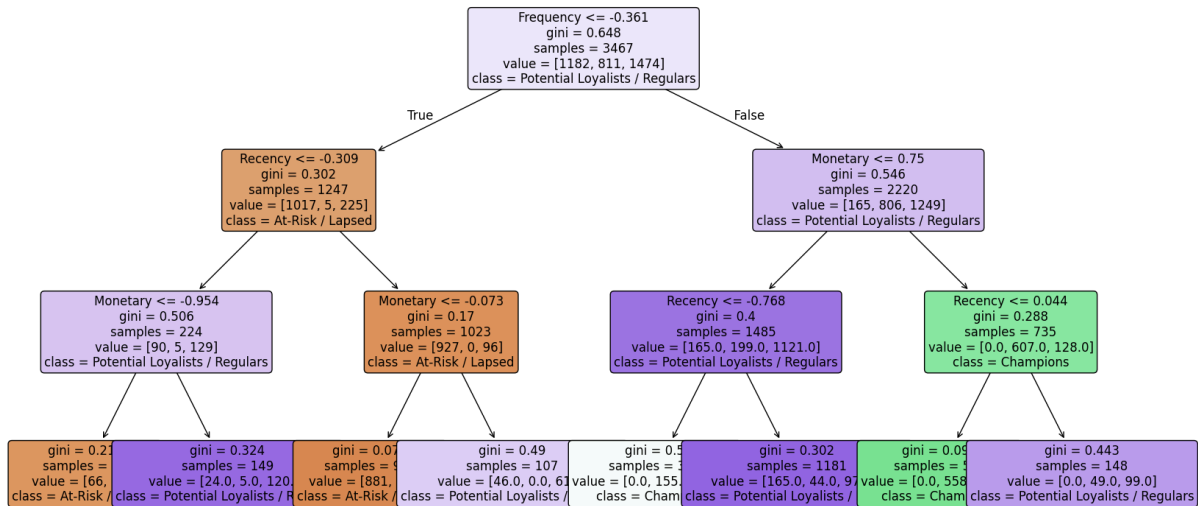
Confusion Matrices for Model Validation



--- Logistic Regression Classification Report ---

	precision	recall	f1-score	support
At-Risk / Lapsed	1.00	1.00	1.00	296
Champions	1.00	1.00	1.00	203
Potential Loyalists / Regulars	1.00	1.00	1.00	368
accuracy			1.00	867
macro avg	1.00	1.00	1.00	867
weighted avg	1.00	1.00	1.00	867

Decision Tree Rules for Customer Segments



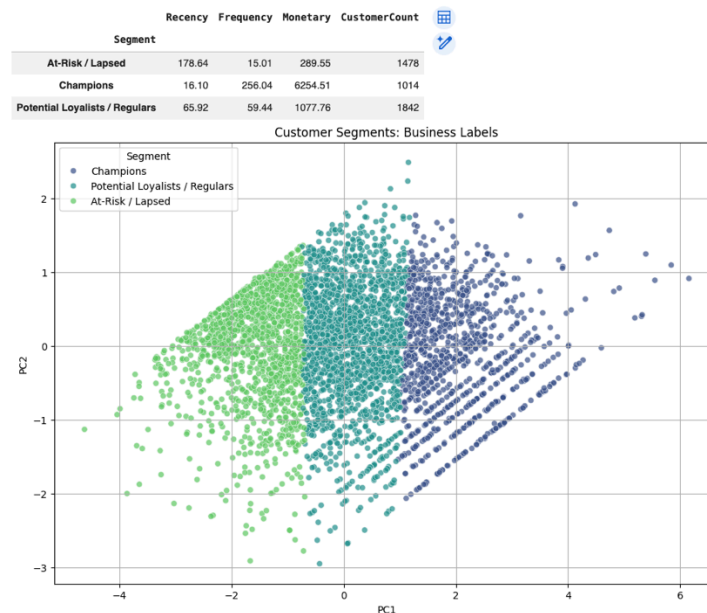
4. Product Affinity and Segment Insights

To bridge the gap between data and business, we analyzed product affinity for each cluster. This revealed that not only do clusters differ in how much they spend, but also in what they buy.

This deep-dive into the "Description" and "StockCode" fields allowed us to characterize the three segments:

- **Champions** - Our most valuable group, buying high-margin items like ceramic jars and decorative ornaments in bulk.
- **Regulars** - Consistent shoppers with mid-range LTV, showing high affinity for utility items and stationery.
- **Lapsers** - Customers at high risk of churn, primarily purchasing small impulse items like balms and purses.

Figure 3: Product Concentration by Cluster



Recommended Products by Segment

To maximize engagement and revenue, the following product categories are prioritized based on the unique behavioral profile of each segment:

- **Champions:** High-end decorative sets and bulk storage solutions to maintain their high monetary value and high-margin contributions
 - 23843 PAPER CRAFT , LITTLE BIRDIE
 - 84077 WORLD WAR 2 GLIDERS ASSTD DESIGNS
 - 85099B JUMBO BAG RED RETROSPOT
 - 22197 POPCORN HOLDER
 - 84879 ASSORTED COLOUR BIRD ORNAMENT

- **Potential Loyalists / Regulars:** "Starter bundles" comprising stationery and utility home goods to leverage their frequent purchasing habits and encourage repeat transactions.

23166	MEDIUM CERAMIC TOP STORAGE JAR
84077	WORLD WAR 2 GLIDERS ASSTD DESIGNS
21212	PACK OF 72 RETROSPOT CAKE CASES
85123A	WHITE HANGING HEART T-LIGHT HOLDER
84879	ASSORTED COLOUR BIRD ORNAMENT

- **At-Risk / Lapsed:** Small, low-friction "loss leader" items such as accessories and balms to provide an attractive, low-cost incentive for re-engagement.

18007	ESSENTIAL BALM 3.5g TIN IN ENVELOPE
17003	BROCADE RING PURSE
22197	SMALL POPCORN HOLDER
84077	WORLD WAR 2 GLIDERS ASSTD DESIGNS
15036	ASSORTED COLOURS SILK FAN

Project Conclusion & Strategy

Our analysis reveals three distinct customer behaviors, allowing us to pivot from generic marketing to data-driven, actionable strategies:

- **Champions (High Frequency/Value):** These shoppers drive the majority of our revenue and have the highest engagement.
 - **Action:** Implement an exclusive VIP loyalty program and provide early-access to new product launches. Since they already favor high-margin decor, use data-backed personalized email campaigns featuring cross-sell opportunities for "complementary" premium items (e.g., matching accessory sets for their recent purchases).
- **Potential Loyalists / Regulars (Moderate Activity):** This segment represents our greatest growth opportunity. They have consistent habits but haven't reached "Champion" status.
 - **Action:** Execute a "Upgrade to VIP" campaign. Create tiered "starter bundles" that combine their favorite utility items with high-margin products currently popular with the Champions segment. This leverages their existing purchase affinity while gently nudging them toward higher basket sizes.
- **At-Risk / Lapsed (Low Recency/Frequency):** This is our largest segment, and their inactivity poses a significant retention risk.
 - **Action:** Initiate a "Win-Back" automation sequence. Use "hook" products identified in our affinity analysis—specifically small, low-friction items like balms and purses—as low-cost incentives. Offer a time-sensitive, targeted discount on these specific items to trigger a low-barrier re-engagement purchase and recapture their attention.

Critical Limitations Analysis

No model is perfect, and it's important to be transparent about where our analysis might face challenges.

1. **The Reality of "Big" Data:** In retail, we naturally see an 80/20 split—a small, elite group of "Champions" and a much larger group of infrequent shoppers. We need to be careful that we don't let this skew our marketing, prioritizing volume over true customer value.
2. **Snapshot vs. Real-Time:** This analysis is a look back at a specific window of time. If we try to apply these exact rules to live, incoming data without adjusting our "snapshot" settings, we risk "cheating" by using future information to make past predictions.
3. **Location Matters:** Our data is heavily UK-centric. If we deploy this model internationally, we might misunderstand unique local shopping habits. To fix this, we need to normalize the data by region so we aren't judging an international shopper by UK standards.